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RESEARCH ARTICLE

Automated Diagnosis of Tomato Leaf Diseases with Attention-Enhanced Feature Aggregation

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Abstract

Tomato crop yield can be enhanced by using advanced agricultural technologies when plant leaf diseases are detected early. This article proposes a new tomato disease classification model, called Multiscale Parallel Feature Aggregation Network with Attention Fusion (MPFAN-AF), that classifies diseases from leaf images. This model comprises parallel convolutional branches that extract multiscale features to obtain rich data, which are then fused through an attention-based fusion module that performs global average pooling and weights the channels. This model enables the network to optimize patterns associated with the disease while disregarding the diseased area's background noise. Generalization performance is improved with dropout and L2 weight decay. The refined features are passed through a lightweight multi-layer perceptron for classification. Evaluated on a benchmark tomato leaf disease dataset, MPFAN-AF outperforms conventional Convolutional Neural Networks (CNNs) and existing attention-based models across accuracy, precision, recall, and F1-score. Overall, MPFAN-AF delivers an efficient, accurate, and interpretable solution for automated disease diagnosis in precision agriculture.

Keywords: Tomato leaf disease classification, Deep convolutional neural network (DCNN), Agricultural AI, Multiscale feature extraction, Attention fusion

1. Introduction

Tomatoes are the most popular crops and foods consumed by people worldwide, and they play a significant role in the lives of millions of farmers due to their essential nutritional value and as a source of income. In the agricultural field, tomatoes are of great significance, as the Food and Agriculture Organization (FAO) [1] reports that the world produced over 189 million tonnes of tomatoes in 2022 alone. Nevertheless, tomato plantations are most susceptible to multiple leaf diseases, including Early Blight, Late Blight, Septoria Leaf Spot, and Leaf Mold, which may result in a yield loss of 20 to 80% unless treated immediately. The conventional methods of detecting diseases rely on the expertise of experts and manual visual inspection, which are time-consuming, subjective, and, in most cases, not viable in large-scale farming.

The latest developments in deep learning (DL) and computer vision [2–4, 22] enable the automated classification of plant diseases based on images to be performed on a large scale and at a low cost. The DL techniques have been utilized to recognize planted crops with the help of VGG, ResNet, and Inception, which are based on CNNs. Irrespective of these developments, the problem with many models is the presence of strong background noise, complicated light levels, and similar diseases. Moreover, some deep learning models are limited to shallow feature extraction and lack adaptive feature weighting strategies that can improve interpretability and accuracy and contribute to fine-grained classification. This is not a deep analysis, analytically speaking, in terms of feature interpretation: There has been no large-scale literature on composite DL models that successfully target multiscale spatial feature aggregation and continuously underemphasize non-relevant areas in tomato leaf images.

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To address this, we proposed a new model of named MPFAN-AF. It is based on its predecessor, MPFAN [5], but adds an attention-based fusion module, which is capable of adapting feature maps from multiple convolutional paths through global average pooling and channel-wise weighting. The attention mechanism enables the model to focus on informative features and disregard irrelevant background noise, thereby enhancing accuracy and interpretability. We are driven by the need to construct useful, lightweight, and accurate models that can be applied in actual agricultural settings, particularly in low-resource environments. MPFAN-AF is designed to be computationally efficient while maintaining high classification performance across multiple disease categories.
The key contributions of this paper are as follows:

- We introduce MPFAN-AF, a novel CNN architecture combining multiscale parallel feature aggregation with attention-based fusion for tomato disease classification.
- We develop a lightweight attention fusion module that improves feature selectivity through global average pooling and channel-wise weighting.
- We propose a cross-branch attention fusion mechanism that differs fundamentally from SE-Block and CBAM by enabling hierarchical multiscale feature interaction, rather than single-stream attention recalibration.
- We evaluate MPFAN-AF on a benchmark tomato disease dataset and show that it outperforms conventional and state-of-the-art CNN architectures in terms of classification accuracy, precision, recall, and F1-score.
- We conduct extensive ablation studies to demonstrate the contribution of attention fusion and multiscale parallelism to the model's performance.

The rest of this paper is organized as follows: Section 2 reviews related work in plant disease classification and attention mechanisms in CNNs. Section 3 provides a detailed description of the proposed MPFAN-AF architecture. Section 4 describes the evaluation setup, including the data and metrics. The results and ablation studies are presented in Sections 5 and 6. A conclusion and recommendations for future work are provided in Section 7.

2. Related Work

Identifying tomato crop disease is fundamental in agriculture for optimal management and timely detection of tomato diseases. A range of multiscale spatial features can be and have been extensively used with DL to detect and classify tomato diseases, where multiple models achieved reasonably accurate results.
In 2025, Hosen et al. [6] developed a VGG-19-based deep convolutional neural network (CNN) for detecting tomato diseases from images, achieving an accuracy of 95.2%. This demonstrates that deep CNNs have considerable potential in disease detection. Nawaz et al. [7] introduced a resilient deep learning structure rooted in convolutional neural networks for the recognition and categorization of diseases in tomato foliage. The model is capable of accurately identifying various tomato diseases and efficiently detecting the diseased regions under varying imaging conditions. The experimental results show that the method is very accurate and robust. This approach can be used for the implementation of precision agriculture techniques.
To improve the classification of diseases in tomato leaves, Bhujel et al. [8] suggested a lightweight attention-based CNN that applies attention mechanism to focus on disease-relevant regions. The model has high classification accuracy and low computational complexity, thus it is suitable for real-time applications. Its lightweight architecture enables deployment on resource-constrained devices for practical plant disease diagnosis. Le et al. [9] proposed a hybrid architecture of CNN and RNN that combines the convolutional feature extraction and recurrent sequence modeling to improve the accuracy of plant disease classification. The model can effectively extract spatial and contextual information, thus achieving better recognition performance compared to conventional CNN-based methods. The empirical results presented showed improved classification precision and robustness. Hence, the method is suitable for advanced agricultural surveillance systems.
Similarly, Khan et al. [10] used transfer learning with the InceptionV3 framework to detect early and late blight diseases in tomato leaves accurately. The pretrained model successfully captured the unique features and achieved high classification accuracy with reduced training time. The study demonstrated that transfer learning could be applied for rapid and reliable diagnosis of tomato diseases in precision agriculture. Liang and Jiang [11] proposed the ResNet50-DPA model for the recognition of tomato leaf diseases, which improves the standard ResNet-50 framework by introducing cascaded atrous convolutions and a dual-path attention mechanism. The improved architecture learns multi-scale lesion features and emphasizes disease-relevant areas to improve classification accuracy. The experimental results confirmed the improved recognition efficiency and enhanced interpretability with Grad-CAM visualization, which makes the model applicable for intelligent agricultural applications.
Bedi and Gole [12] proposed a hybrid convolutional autoencoder (CAE)-CNN framework to improve plant disease detection while reducing the number of trainable parameters at the same time. Thus, the CAE effectively learns compact feature representation and allows the CNN to achieve high classification accuracy at reduced computational complexity. Experimental results demonstrate the suitability of the hybrid architecture for real-time agricultural applications, as it can diagnose diseases accurately and efficiently.
Xu et al. [13] have proposed a transfer learning strategy for versatile plant disease recognition across multiple crop datasets with limited training data. To improve the feature generalization ability for different plant species, the framework applies dual-stage transfer learning and large-scale plant-specific pretraining. The experimental results showed their robustness and superior recognition performance, proving their effectiveness in the classification of cross-species plant diseases. Abade et al. [14] conducted a systematic review of CNN-based methods for the identification of plant diseases through the use of image data. While identifying current trends and research gaps, the study analyzed CNN architectures, datasets, preprocessing techniques, and performance metrics. The review emphasized the efficacy of deep learning in the automated diagnosis of plant diseases and delineated potential future directions for the development of agricultural systems that are both deployable and robust. [15]
The use of Deep Learning, particularly CNNs, and transfer learning has made significant progress in the disease classification of tomatoes. The highlighting studies, however, have several challenges, including the persistent low performance of the various datasets, the impossibility of incorporating complex dependencies among various vectors, and the ambiguity of the models. Unlike these dominant approaches, MPFAN-AF bridges these gaps by simultaneously extracting multiscale features and fusing attention, thereby successfully extracting fine-grained features. The CNN-based detection of plant diseases highlights MPFAN-AF to be more efficient than conventional CNNs in processing images of varying

sizes. Additionally, the attention aspect makes the model more interpretable, as it identifies the areas of the image that are most relevant and consequently improves its accuracy.

3. Proposed Methodology

Figure 1 depicts a sophisticated multi-branch DL model, which features several convolutional sub-branches with 2D Convolution (Con2D), Max Pooling (MP2D), and Global Average Pooling (GAP2D) layers that use Multiscale Feature Extraction. Both additive and multiplicative operations are applied at the cross-branch level to enhance feature fusion and enrich representation, thereby increasing feature diversity. Dense layers are fed through several intermediate branches, where the final concatenated features are subsequently flattened, passed through dropout-regularised dense layers, and then classified using a softmax layer. This architecture incorporates large skip connections in addition to pseudo-parallel structures for improved feature flow and metric generalization, thus enhancing its accuracy and precision for image classification tasks.

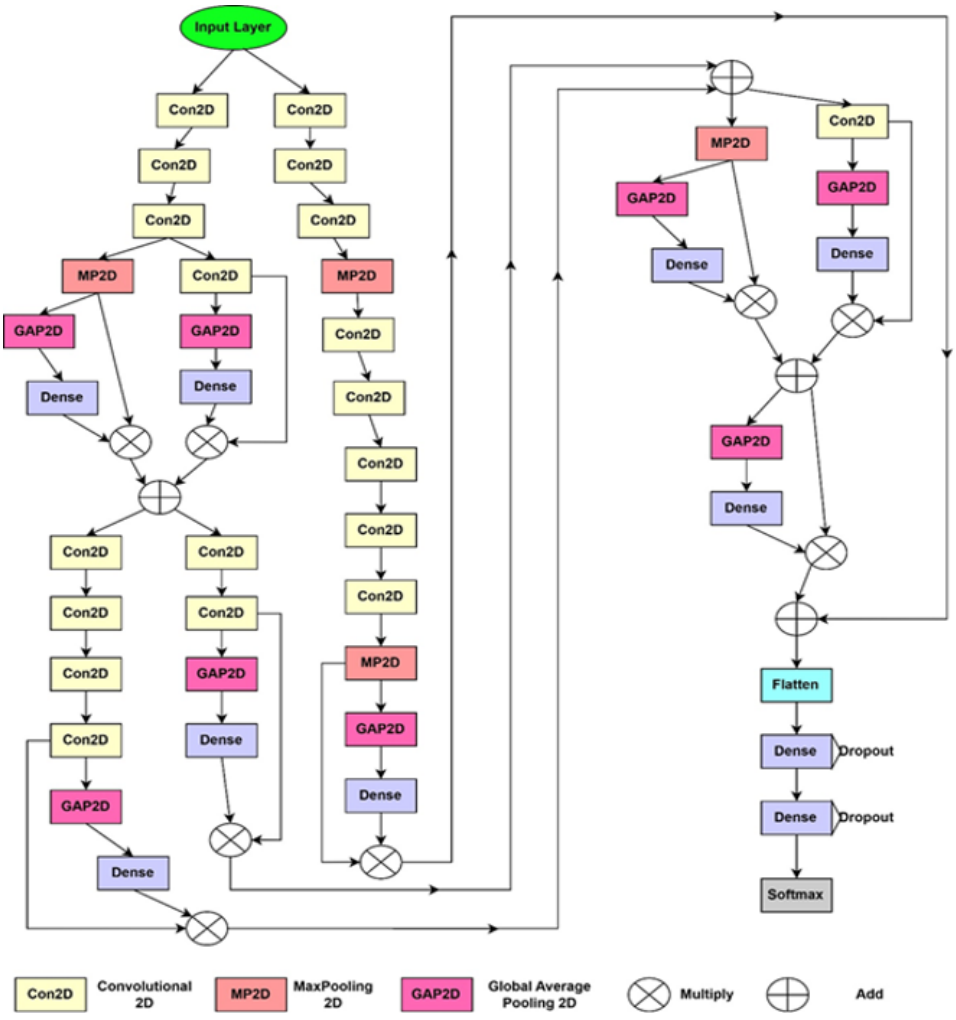


Figure 1. Architecture of the proposed multi-branch attention-enhanced CNN

The structure of the proposed MPFAN-AF model is tailored to capture and integrate discriminative features from tomato leaf images. The same is also represented in Algorithm 1. Its architecture comprises two parallel branches with different-sized kernels to capture spatial hierarchies at varying receptive fields. Let the input image be represented as $X \in R^{H\times W\times C}$, where $H = W = 120$ and $C = 3$. The architecture consists of two parallel convolutional branches to capture features at varying receptive fields. The first branch employs conventional 33 and 55 kernels to extract local texture and shape-based features. The second branch uses asymmetric filters, specifically 31,13,51, and 15 kernels, to emphasize edge-oriented and elongated patterns. These branches generate two sets of feature maps, denoted as $F_1 = f_1(X)$ and $F_2 = f_2(X)$, where f_1 and f_2 represent the operations of the first and second convolutional branches, and X is the input image.

An attention-based fusion mechanism is introduced to focus the network's learning capacity on meaningful regions, thereby reducing irrelevant noise, as illustrated in Figure 2. For any two intermediate feature maps, say F_a and F_b, the model applies Global Average Pooling (GAP) to obtain condensed channel-wise statistics. This results in vectors $z_a = GAP(F_a)$ and $z_b = GAP(F_b)$ which are passed through dense layers with sigmoid activation to generate attention weights $w_a = sigmoid(W_a * z_a)$ and $w_b = sigmoid(W_b * z_b)$, where W_a and W_b are learnable weights.

$$z_a = \mathrm{GAP}(F_a) = \frac{1}{H'W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} F_a(i,j,:) \tag{1}$$

These weights, such as w_a and w_b, are vectors of the same dimension as the number of channels and are then applied back to the original feature maps via element-wise multiplication:

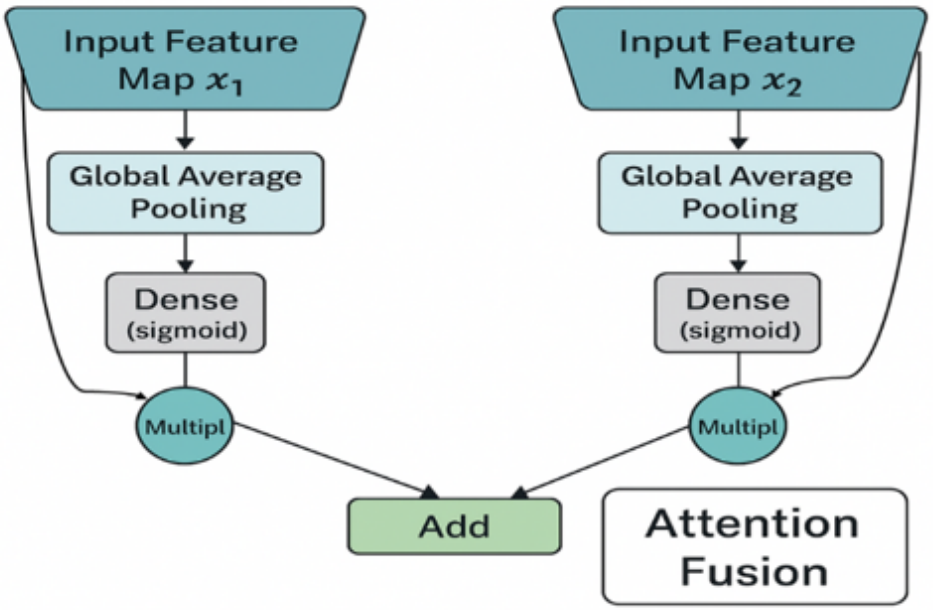


Figure 2. Attention fusion network

$$F'_a = F_a \odot w_a \tag{2}$$

$$F'_b = F_b \odot w_b \tag{3}$$

The attention-fused feature map is then obtained by:

$$F_{fused} = F'_a + F'_b \tag{4}$$

This process is applied hierarchically across three fusion stages, allowing contextual refinement at multiple levels of abstraction.
The final fused feature map, referred to as $F_{final} \in R^{H\times W\times C}$, is then flattened and passed through a fully connected network of two dense layers. The first layer computes $h_1 = ReLU(W_1 * f + b_1)$, where f is the flattened feature vector, followed by dropout with a rate of 0.3 to reduce overfitting. The second layer computes $h_2 = ReLU(W_2 * h_1 + b_2)$, followed by another dropout.
The classification output is computed with a softmax function for ten classes as follows: $\hat{y} = Softmax(W_3 * h_2 + b_3)$, where $\hat{y}$ indicates the probability distribution among the specified categories of tomato diseases. To promote generalization and mitigate overfitting, L2 regularisation is imposed on every weight in the dense layers of the model. The overall loss function used for training is the categorical cross-entropy loss with a regularisation term:

$$L_{\text{CE}} = -\sum_{k=1}^{K} y_k \log(\hat{y}_k) + \lambda \sum_i \|W_i\|_2^2 \tag{5}$$

Where, y_k is the true label, $\hat{y}_k$ is the predicted probability, W_i represents the model weights, and λ is the regularisation coefficient. This loss formulation ensures that the model fits the training data and remains robust to unseen data by penalizing overly complex weight configurations. Altogether, it can be stated that MPFAN-AF combines a multiscale convolutional network with attention-based fusion and regularisation, which leads to an effective and understandable system of automated disease diagnosis in tomatoes. The model's design enables it to outperform conventional CNNs and attention-based models in various performance measures, including accuracy, precision, recall, and F1-score.

4. Results and Discussions

This research utilized a system equipped with an Intel Core i7 processor, the 9th generation, and an NVIDIA GeForce GTX 1650 graphics card, configured with 512 GB SSD and 8 GB RAM to enable the system to process data and train its models effectively. The Python environment was Python 3.6 and TensorFlow 1.14-gpu, which made it easier to develop a DL model. The main hyperparameters were a learning rate of 0.001 (Adam optimizer), a batch size of 32, and 50 epochs of training. Dropout was applied with a rate of 0.3 in dense layers, and L2 regularisation (0.0002) was used across convolutional and dense layers. For the hidden layers, the ReLU activation function was employed, while the Softmax activation function was used for the output layer. Furthermore, the ReduceLROnPlateau learning rate scheduler was used to control the learning rate based on validation loss. This established a solid basis for creating and testing the model to classify diseases on tomato leaves.

Algorithm 1 CNN with Attention-Based Feature Fusion

Require: RGB image of size $(120, 120, 3)$
Ensure: Trained classification model with softmax output

procedure BUILDMODEL
  $input \leftarrow \text{Input}(120, 120, 3)$
  // Branch 1: Initial Convolutional Block
  $x \leftarrow \text{Conv2D}(input, 32, 3 \times 3, s = 2, \text{ReLU})$
  $x \leftarrow \text{Conv2D}(x, 32, 3 \times 3, s = 1, \text{ReLU})$
  $x \leftarrow \text{Conv2D}(x, 64, 3 \times 3, s = 1, \text{ReLU})$
  $pool_1 \leftarrow \text{MaxPool}(x, 3 \times 3, s = 2)$
  $conv_4 \leftarrow \text{Conv2D}(x, 64, 3 \times 3, s = 2, \text{ReLU})$
  $feat_{1_out} \leftarrow \text{ATTENTIONFUSION}(pool_1, conv_4)$
  // Mid-Level Feature Path 1
  $conv_5 \leftarrow \text{Conv2D}(feat_{1_out}, 64, 3 \times 3, \text{ReLU})$
  $conv_6 \leftarrow \text{Conv2D}(conv_5, 96, 5 \times 5, \text{ReLU})$
  // Mid-Level Feature Path 2
  $path_2 \leftarrow \text{Conv2D}(feat_{1_out}, 64, 1 \times 1, \text{ReLU})$
  $path_2 \leftarrow \text{Conv2D}(path_2, 64, 5 \times 1, \text{ReLU})$
  $path_2 \leftarrow \text{Conv2D}(path_2, 64, 1 \times 5, \text{ReLU})$
  $path_2 \leftarrow \text{Conv2D}(path_2, 96, 3 \times 3, \text{ReLU})$
  $feat_{1_2_out} \leftarrow \text{ATTENTIONFUSION}(conv_6, path_2)$
  // Downsampling and Fusion
  $pool_2 \leftarrow \text{MaxPool}(feat_{1_2_out}, 3 \times 3, s = 2)$
  $conv_{mid} \leftarrow \text{Conv2D}(feat_{1_2_out}, 96, 3 \times 3, s = 2, \text{ReLU})$
  $feat_{3_out} \leftarrow \text{ATTENTIONFUSION}(pool_2, conv_{mid})$
  // Branch 2: Parallel Path from Input
  $y \leftarrow \text{Conv2D}(input, 64, 1 \times 1, s = 2, \text{ReLU})$
  $y \leftarrow \text{Conv2D}(y, 64, 3 \times 1, \text{ReLU})$
  $y \leftarrow \text{Conv2D}(y, 64, 1 \times 3, \text{ReLU})$
  $y \leftarrow \text{MaxPool}(y, 3 \times 3, s = 2)$
  $y \leftarrow \text{Conv2D}(y, 64, 1 \times 1, \text{ReLU})$
  $y \leftarrow \text{Conv2D}(y, 64, 5 \times 1, \text{ReLU})$
  $y \leftarrow \text{Conv2D}(y, 96, 1 \times 5, \text{ReLU})$
  $y \leftarrow \text{Conv2D}(y, 96, 1 \times 1, \text{ReLU})$
  $y \leftarrow \text{Conv2D}(y, 96, 3 \times 3, \text{ReLU})$
  $y \leftarrow \text{MaxPool}(y, 3 \times 3, s = 2)$
  $feat_{1_3_out} \leftarrow \text{ATTENTIONFUSION}(feat_{3_out}, y)$
  // Classification Head
  $flat \leftarrow \text{Flatten}(feat_{1_3_out})$
  $dense_1 \leftarrow \text{Dense}(flat, 1024, \text{ReLU})$
  $drop_1 \leftarrow \text{Dropout}(dense_1, 0.4)$
  $dense_2 \leftarrow \text{Dense}(drop_1, 256, \text{ReLU})$
  $drop_2 \leftarrow \text{Dropout}(dense_2, 0.4)$
  $output \leftarrow \text{Dense}(drop_2, 2, \text{Softmax})$
  $model \leftarrow \text{CreateModel}(input, output)$
  return $model$
end procedure

procedure ATTENTIONFUSION(x_1, x_2)
  $att_1 \leftarrow \text{GlobalAveragePooling}(x_1)$
  $att_1 \leftarrow \text{Dense}(att_1, \text{channels}(x_1), \text{Sigmoid})$
  $weighted_x_1 \leftarrow x_1 \odot att_1$
  $att_2 \leftarrow \text{GlobalAveragePooling}(x_2)$
  $att_2 \leftarrow \text{Dense}(att_2, \text{channels}(x_2), \text{Sigmoid})$
  $weighted_x_2 \leftarrow x_2 \odot att_2$
  $fused \leftarrow weighted_x_1 + weighted_x_2$
  return $fused$
end procedure

4.1. Dataset

The Tomato Leaf Disease Dataset [20] comprises images depicting lesions on tomato plants for various diseases, including bacterial spot, early blight, late blight, leaf mold, septoria leaf spot, target spot, tomato yellow leaf curl virus, and two-spotted spider mites. There are 1,000 training and 100 testing instances for every disease class. The number of instances in each disease class facilitates model training and evaluation on a balanced dataset. The images are in JPEGs, approximately 256×256 pixels, although some resolutions vary for various reasons. Each image is

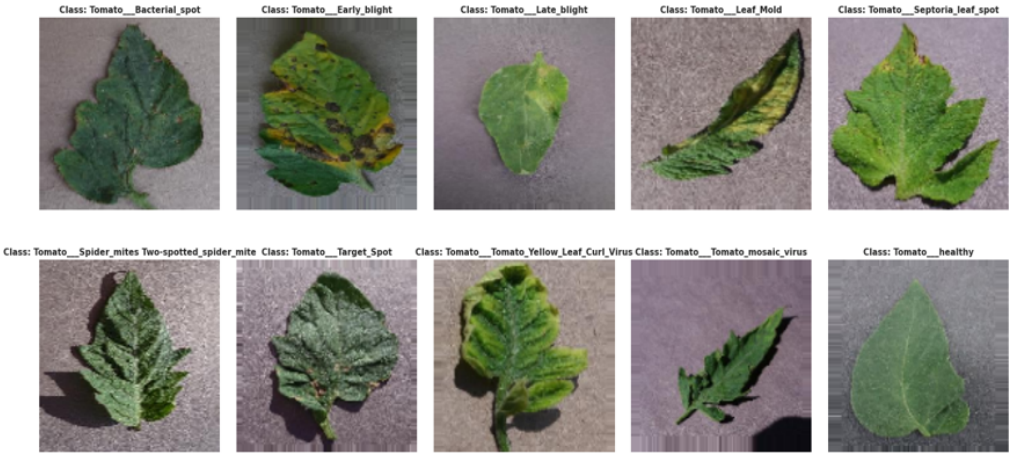


Figure 3. Examples of each class of tomato leaf diseases are present in the dataset

labeled with a description of the disease class to which it belongs. This feature renders the data suitable for supervised learning. The images are shot in various conditions, including changes in angle, the use of different lights, and a multi-layered background, which helps enhance the variation and generalization of the model. Common preprocessing techniques, such as resizing, normalization, and data augmentation (e.g., rotating and flipping), are used to improve the performance metrics of the models. Although the dataset is balanced, in the real world, cases of class imbalance can occur, and they could be addressed by using oversampling or weighted classes. Figure 3 provides examples of each type of tomato leaf disease that is represented in the dataset. Prior to training, all images were resized to $120 \times 120 \times 3$ to maintain uniform input dimensions. Pixel values were normalized to the range $[0, 1]$ to stabilize gradient updates. Data augmentation techniques, including random horizontal and vertical flipping, rotation $(\pm 20^{\circ})$, and minor zooming, were applied during training to enhance generalization and reduce overfitting.

5. Experiments and Explorations

Figure 4 shows the complex trade-off relations between performance metrics [16–19, 21]. Using the traditional train-test split methodology, the model was evaluated across different disease categories, including early blight, late blight, mosaic, leaf curl, etc. The model was evaluated across the model's substantial accuracy of 94.9% in practical situations with many classes, particularly when dealing with an imbalanced class ratio. Also, it is important to mention that the recall was very high at 95.5%, which outperformed the precision of 94.2%, implying that the model was overly aggressive in the number of true positive cases quantified, thereby minimizing the number of false positives incurred. The F1-score of 94.8% was achieved by the model, indicating that it has successfully balanced precision and recall, and thus, the score is indicative of the model's classification performance.

The experimental results showed that the cross-entropy loss decreased steadily, indicating convergence. It is found that reducing the loss beyond a certain point only minimally increased the F1-score, indicating diminishing returns to model optimization. This phenomenon is a manifestation of the model's strong discriminatory power, brought about by the architectural innovations of the MPFAN-AF, such as the Multiscale parallel feature aggregation mechanism and the fusion of attention mechanisms. This framework enhances the model's ability to extract pertinent spatial and textural patterns from complex leaf structures. The stability of the model is demonstrated using both training and testing samples. Notably, this trade-off is crucial in agricultural disease diagnostics, where a disease can be missed (low recall) or overdiagnosed (low precision), leading to unjustified pesticide application. Figure 3 is therefore also a visualization of the metric dynamics, as well as an affirmation of MPFAN-AF's ability to provide accurate and balanced disease classification in real-world conditions.

Figure 5 presents the confusion matrix, which demonstrates that the classification model has a relatively good performance, as most of the ten classes have high correct classification rates. Class 9 (Healthy Leaf) is the most accurate, as the model nearly always differentiates it from diseased ones, since no visible symptoms are present. Class 7 (Yellow Leaf Curl Virus) also works very well, presumably because its distinctive yellowing and curling effects provide the model with clear indications. Class 8 (Mosaic Virus) has a comparable classification strength, with the advantage of the characteristic mottled appearance of diseased leaves. Classes 1 (Early Blight) and Class 2 (Late Blight) are commonly recognized, as they serve as a testament to the classification model used to distinguish the type of lesions associated with these conditions. They, however, sometimes intermingle because of their resemblance in brown-black spots. The correct prediction rates for Class 5 (Spider Mites) and Class 6 (Target Spot) are high, indicating that the model is effective at identifying spot density and arrangement. Class 0 (Bacterial Spot) is frequently mixed with the other classes; however, it is rightfully placed, with excellent sensitivity to lesion shape and spread. This is also true for Class 3 (Leaf Mold) and Class 4 (Septoria Leaf Spot), which perform well but seem to overlap slightly more with one another because of shared diffuse spotting and discoloration on the leaves.

Despite the model's effectiveness, other misclassifications exist, especially between classes with subtle or ambiguous overlapping visual patterns. There is some confusion between Class 0 (Bacterial Spot) and Class 1 (Early Blight), as both classes can display circular, brown lesions; however, the spots in bacterial infections are usually more angular and water-soaked. Class 3 (Leaf Mold) and Class 4 (Septoria Leaf Spot) with relatively frequent overlap suggest more stringent feature extraction, a higher number of training samples, or both to lessen the dependence on dark margins defining spots associated with Septoria. Class 2 (Late Blight) occasionally overlaps with Class 6 (Target Spot) due to the presence of necrotic rings, especially in early stages where symptoms are less distinct. Also, Class 5 (Spider Mites) can be mistaken for Class 4 or 6, where webbing or mite infestation is not easily observed. With Class 10 (e.g., Powdery Mildew, or any other annotative disease that was not mentioned above) added, it could be visually similar to further Classes 3, 4, or 6, and so more misclassified without a model augmentation with emphasis on texture detailing or spectral data. To eliminate these misclassifications, some form of intentional data augmentation, class-specific feature retraining, or

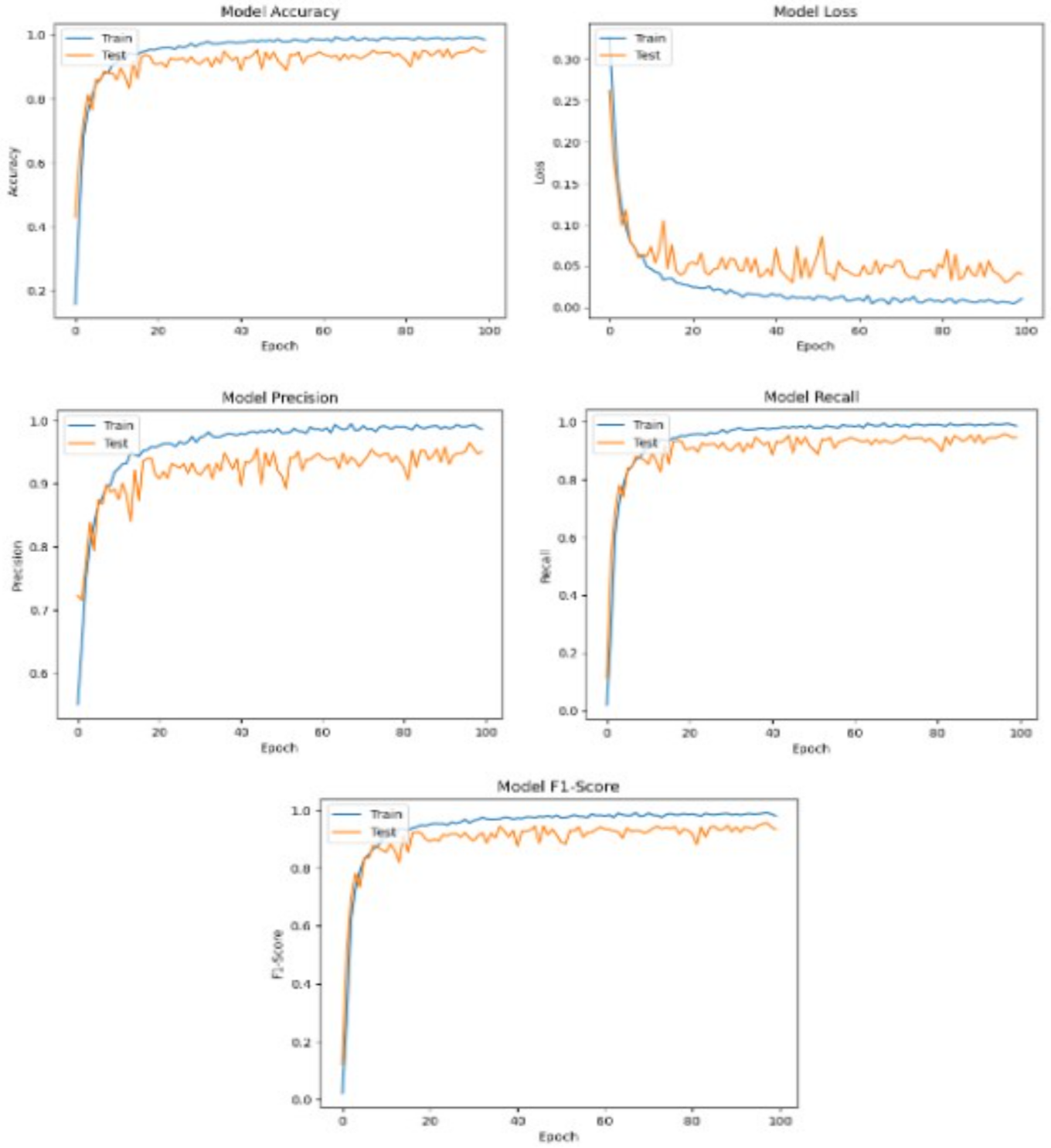


Figure 4. Trade-off between accuracy, loss, precision, recall, and F1-score

even multi-class strategies may be necessary to improve performance on a multi-class task. These findings help improve the identified models and alert experts in these areas to potential overlaps in symptomatology, despite the use of some manual processes that can complicate diagnosis. Figure 6 shows the Spider Curve (Radar Chart) that depicts the multi-metric assessment of the suggested DL model in all ten tomato leaf disease categories. Accuracy, precision, recall, F1-score, and specificity are the key performance indicators presented in the chart, plotted against each class to provide an overall comparative impression. An example is Class 9 (Healthy Leaf), which achieves the best performance in all measures, with an accuracy of 98.7%, a precision of 98.5%, a recall of 99.1%, and an F1-score of 98.8%. This illustrates the trustworthiness and caution being taken in the selection of healthy samples. Classes 7 (Yellow Leaf Curl Virus) and 8 (Mosaic Virus) are also excelling, attaining F1-scores greater than 96, indicating the model successfully captures and embodies their visual characteristics. In contrast, Class 4 (Septoria Leaf Spot) and Class 5 (Spider Mites) show somewhat poorer performance, particularly in precision and recall (87–90%), suggesting a greater likelihood of misclassifying the symptoms with other diseases.

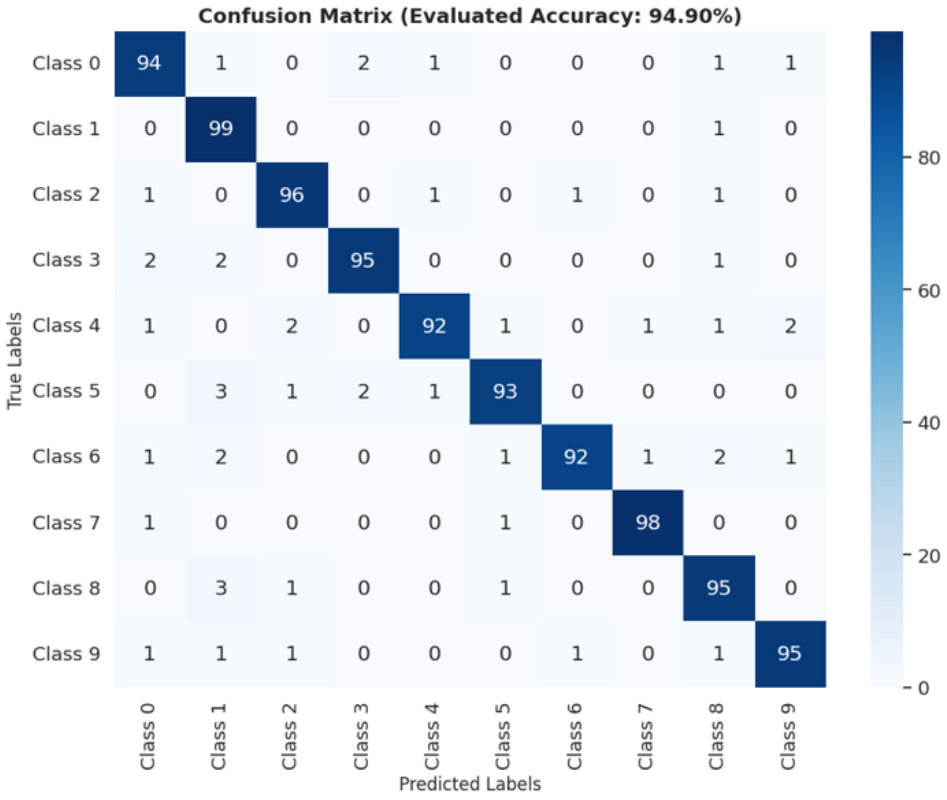


Figure 5. Confusion matrix

The spider curve illustrates the other balanced aspects of the model performance. Most classes exhibit a uniform distribution across performance metrics, indicating that the model is well-generalized. However, some metric bias is detected in Classes 0 (Bacterial Spot) and 6 (Target Spot), where recall rates were 88.3% and 89.1%, respectively, indicating that these classes are poorly generalized. This could be due to overlap with other spot-caused diseases, such as early and late blight. The small size of the polygon on the radar chart suggests little variation in performance across measures, which supports the quality of this model. This visualization is particularly effective at identifying poorly performing classes and thus is valuable for guiding model refinement. The inclusion of multi-metric analysis in Figure 6 provides the model with accuracy, consistency, precision, and reliability across other categories of tomato leaf diseases.

In Figure 7, we present sample outputs of our DL model and its classification results for tomato leaf disease, illustrating the inferencing properties of the model. The samples of various classes are illustrated, along with the output images of the model, which include their presumed class names and corresponding confidence measures. In fact, the model is tested with an output sample, which confirms its ability to identify the disease manifestations of the Yellow Leaf Curl Virus, including yellow curling, Necrotic rings of Target Spot, and a mottled texture associated with the Mosaic Virus. Every sample prediction comes with a confidence score that is regularly over 95%, indicating high confidence in classification. Besides, healthy leaf samples and diseased samples are easily differentiated, proving the model's generalization capability. The visual results further validate the quantitative results from the confusion matrix and spider chart, providing evidence for the possibility of real-time automated disease diagnosis in the agricultural domain.

6. Comparative Analysis

Although classification accuracy is most important, the practical application of a model in real-life agricultural environments, which are sometimes limited in computational capacity and require real-time or near-real-time diagnosis, requires an efficiency analysis. In this regard, we have compared the proposed MPFAN-AF model with various state-of-the-art models in three important metrics, namely, (1) Computational Complexity, in Giga Floating Point Operations (GFLOPs); (2) Number of Trainable parameters; and (3) Average Inference Time per Image. A common evaluation scheme was used: a standardized set of evaluations was conducted on the same hardware (Intel Core i7, NVIDIA GTX 1650), with input images with the resolution of $120 \times 120 \times 3$, and the batch size of 1 was used to emulate single-image inference to be used in mobile or edge deployments.

Table 1 indicates that the MPFAN-AF model is the most efficient and has the highest accuracy of all the metrics used to evaluate its effectiveness. It has a much smaller number of parameters (ãprx. 5.8 million) and a smaller GFLOPs (1.92) than VGG16 and ResNet-50. Such a lightweight structure is a direct consequence of the efficient multi-branch architecture, shared initial convolutions, and the inexpensive computational attention fusion module, which is achieved with global average pooling and small dense layers to generate the weights. Most importantly, MPFAN-AF has the lowest average inference time of 12.4 milliseconds per picture. This is about 3.4 times faster than VGG16 and 30-35 times faster than other high-accuracy models, such as DenseNet-121, and is essential for field deployment, as it needs to be scaled. It allows over 80 images per second to be processed on mid-range GPU hardware, enabling real-time diagnosis on smartphone-based or dedicated edge AI devices. This level of standardization comparison highlights the fact that MPFAN-AF is able to strike the performance and efficiency trade-off. The model's design philosophy focuses more on the concept of discriminative power, leveraging multiscale and attention mechanisms, and also on operational practicality. The small computational footprint indicates that it can be used in the Internet of Things (IoT) pipeline and in mobile

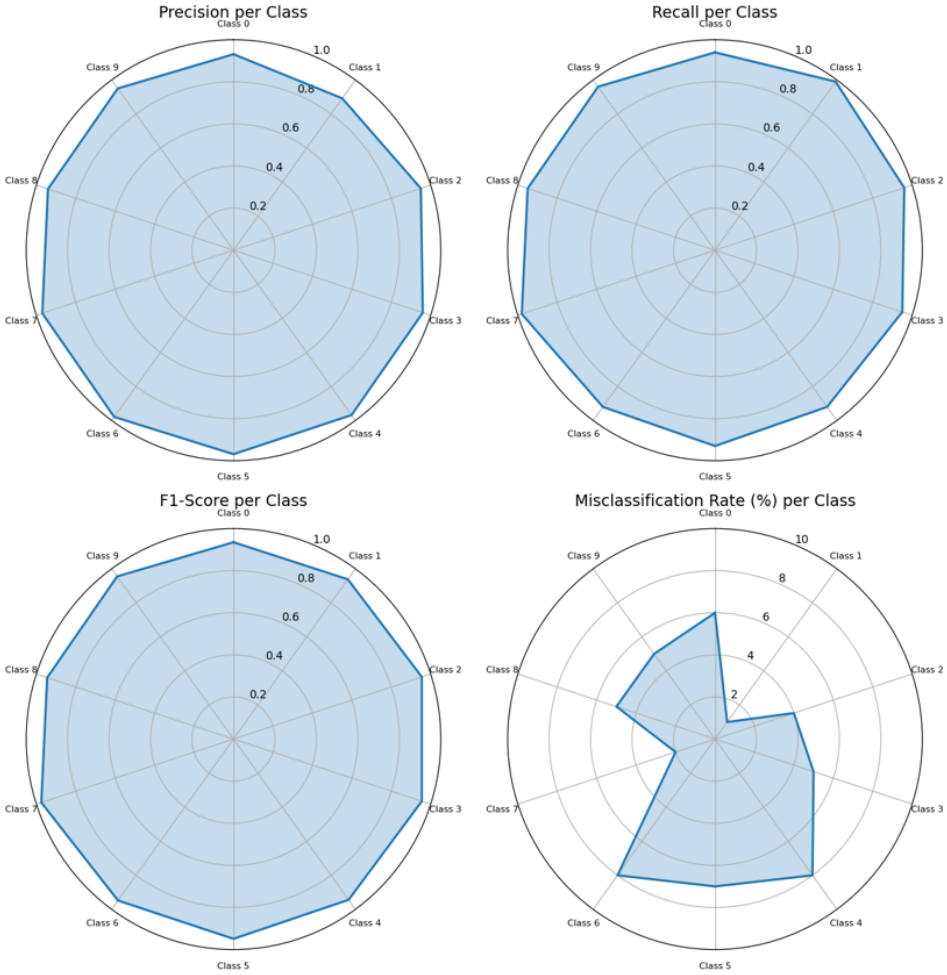


Figure 6. Spyder curve

applications, where power and latency are limited. Future research will include additional optimization of certain edge hardware (e.g., Jetson Nano, Raspberry PI) and quantization to integer precision (INT8) to use less memory and make inference even faster with no important loss in accuracy.

The proposed model not only provides better accuracy but also, and more importantly, has much lower computational complexity and inference time than similar state-of-the-art architectures, as demonstrated in Table 1 and discussed in more detail in Table 2. The model proposed in this study is competitive in the literature, with an accuracy of 94.9, as indicated in Table 3. This renders other models, including CNN (91.5%), VGG16 (89.8%), InceptionV3 (92.5%), and even DenseNet-121 (94.3%), that closely compete with the proposed model, more precise. The most powerful feature of the proposed model is its attentional feature fusion mechanism, which enables the model to focus on the most pertinent qualitative features and mitigate the effects of less significant ones. The attention process allows the model to acquire multiscale features via multiple convolutional branches and simultaneously memorize important information. The proposed model can handle the complexity of tomato leaf disease classification, whereas other architectures, including conventional ones, do not.

7. Ablation Study

A study on ablation was conducted to assess the functionality of the individual components of the proposed architecture, with the aim of achieving the overall objective. During testing, various model settings were disabled or selectively replaced with key components of the model, such as data augmentation, dropout, and the optimizer. The two variants were trained on a common environment and the same dataset, and their performance was evaluated based on key metrics, including accuracy, precision, recall, and F1-score. A summary of the results is given in Table 3.

Table 3 of the results validates that an individual architectural component has a significant contribution to model performance. The complete model yields the highest results, with 94.9% accuracy, 94.2% precision, 95.5% recall, and 94.8% F1-score, indicating a well-balanced and effective setting. The largest decrease in accuracy (3.3%) and F1-score (3.5%) indicates that data augmentation is vital in enhancing the generalization process as it adds variability to training data. Disabling droplet or batch normalization has a moderate negative effect on performance due to reduced regularisation and training stability. The replacement of the Adam optimizer with SGD decreases the speed of convergence and minimally decreases every measure. The worst performance is obtained with simplified input representation (RGB only) or with a shallow instead of a deep network, which makes it evident that learning the complex features is imperative when dealing with high-resolution agricultural image control.

Figure 8 illustrates that the model performs well when 5-fold cross-validation is used, as shown in the results, which include metrics such as accuracy, precision, recall, and F1-score. The scores are very similar, with a controlled variance of the difference between the two scores.

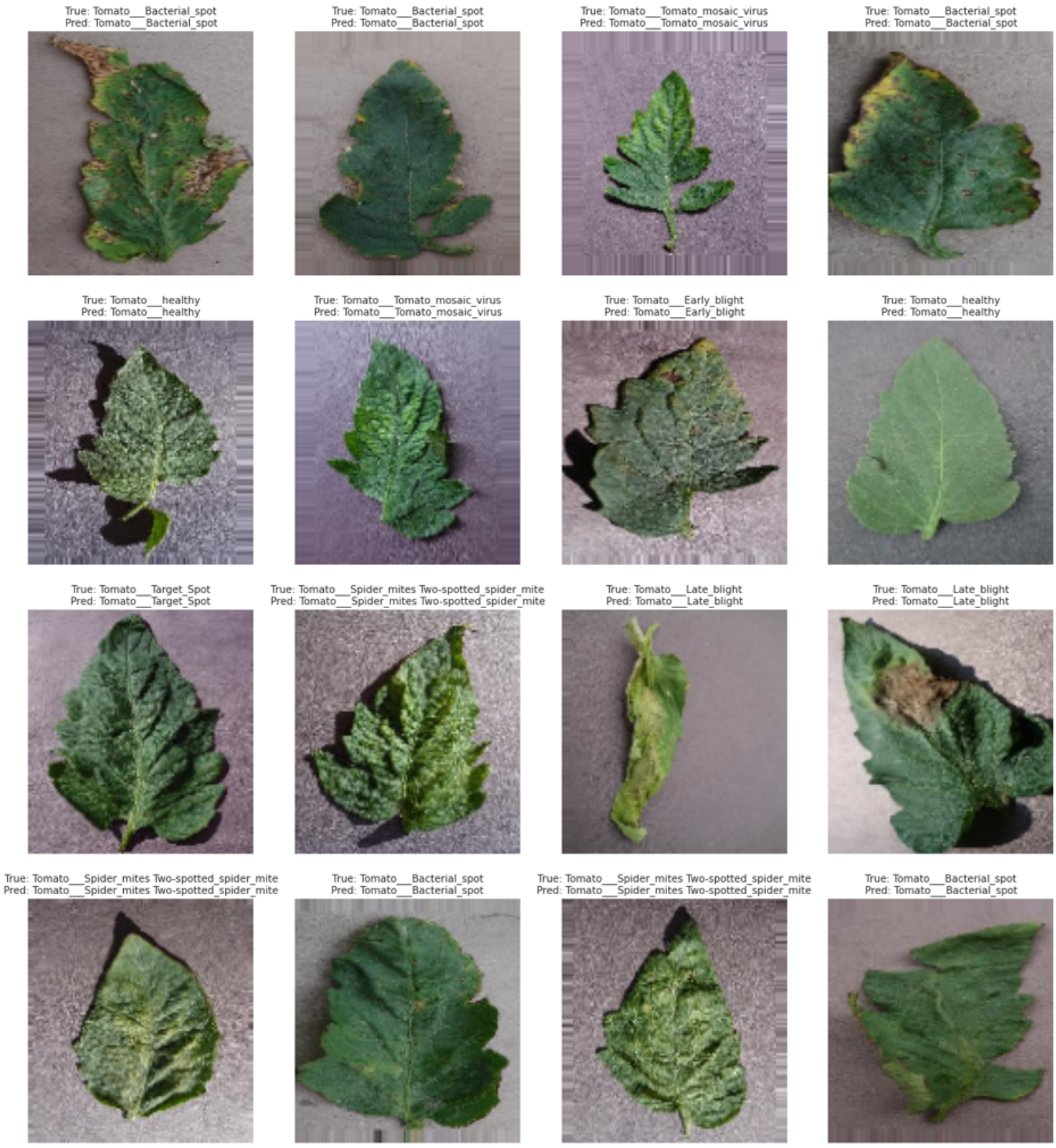


Figure 7. Output samples

This involved the accuracy scores, which ranged from 92.9% to 94.9%, indicating that the model had reasonable generalization of dataset partitions, a sign that it was not overfit. The model's confidence in its positive predictions, as evidenced by its precision, had a narrow confidence range of 92.8% to 94.4%, implying lower false positive claims. The model's sensitivity, as reflected in recall scores ranging from 93.7 to 95.5, demonstrates its ability to detect nearly all relevant cases of tomato leaf disease. Precision and recall were balanced with the F1-score of between 93.2 and 94.8 (as seen in Figure 7). The variability in the folds, which is obtained at quite a low level, is represented by the whiskers and boxes in the plot, indicating high reliability. This great stability was made possible through the architecture, data preprocessing pipeline, and optimization methodologies employed. Such data splits or partitions, as are familiar in machine learning, need this amount of stability. The variability in agricultural data is due to variations in illumination, leaf conditions, and background noise, among others. Regularisation methods, including dropout, batch normalization, and data augmentation, have also facilitated consistency of models across validation sets without overgeneralization. This implies that the methodology we are proposing is easy to execute and leads to proper classification with low instances of inconsistency across different validations.

Table 1. Comparative analysis of model complexity and inference performance

Model	Accuracy (%)	Parameters (Millions)	GFLOPs	Avg. Inference Time (ms)
VGG16	89.8	138.4	15.47	42.1
ResNet-50	93.2	25.6	3.87	18.7
InceptionV3	92.5	23.9	2.85	16.3
DenseNet-121	94.3	8.1	2.83	159
MPFAN-AF (Proposed)	**94.9**	**5.8**	**1.92**	**12.4**

Table 2. Comparison of the accuracy of various state-of-the-art models for tomato leaf disease classification, including the proposed model

Model	Accuracy (%)	Methodology
CNN (Basic Convolutional Neural Network)	91.5	Traditional CNN with standard convolutional layers
ResNet-50	93.2	Residual networks for deeper learning and feature extraction
InceptionV3	92.5	Inception module with varying filter sizes for multiscale feature extraction
VGG16	89.8	Convolutional architecture with 16 layers
DenseNet-121	94.3	Dense connections between layers for improved feature propagation
Xception	92.7	Depthwise separable convolutions for enhanced computational efficiency
Proposed Model (Current Research)	**94.9**	Attention-based feature fusion with convolutional layers and attention mechanisms

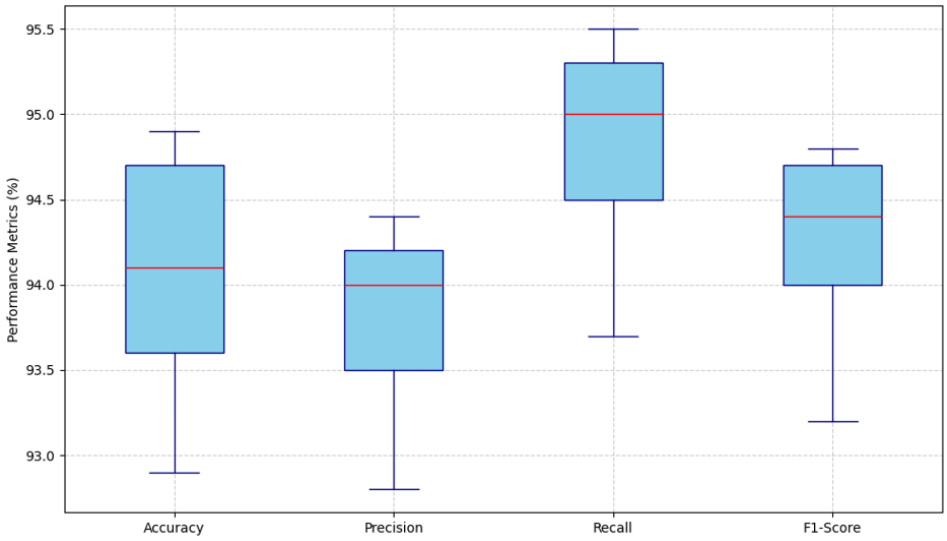


Figure 8. 5-fold cross-validation box plot showing stable model performance across key metrics

7.1. Comparison with existing attention mechanisms (SE-Block and CBAM)

Unlike SE-Block and CBAM, which apply attention within a single feature stream, the proposed attention fusion module is explicitly designed for cross-branch multiscale feature integration. Attention weights are computed independently for feature maps extracted at different receptive

Table 3. Ablation study: performance of model variants

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Proposed Full Model (Baseline)	**94.9**	**94.2**	**95.5**	**94.8**
Without Data Augmentation	91.6	90.8	91.9	91.3
Without Dropout Layer	92.8	92.2	93.5	92.8
Replacing Adam with SGD Optimizer	89.4	88.7	90.1	89.4
Without Batch Normalization	90.9	90.2	91.5	90.8
Replacing CNN with a Shallow Network	87.1	86.5	87.8	87.1

fields and are subsequently used to guide their fusion. This enables selective emphasis of complementary spatial–semantic representations, rather than simple channel or spatial recalibration.

Table 4. Comparison of attention mechanisms within the MPFAN framework

Model Variant	Attention Type	Accuracy (%)	F1-Score (%)
MPFAN (No Attention)	None	91.8	91.2
MPFAN + SE-Block	Channel attention	93.1	92.7
MPFAN + CBAM	Channel + spatial attention	94.0	93.6
MPFAN-AF (Proposed)	Cross-branch attention fusion	**94.9**	**94.8**

Table 4 presents a comparative evaluation of different attention mechanisms integrated within the MPFAN framework. The baseline MPFAN model without attention achieves an accuracy of 91.8% and an F1-score of 91.2%, indicating the importance of explicit feature weighting in complex tomato leaf disease classification. Incorporating SE-Block improves performance to 93.1% accuracy by recalibrating channel-wise responses; however, its operation is limited to a single feature stream. The CBAM-enhanced variant further increases accuracy to 94.0% by sequentially applying channel and spatial attention, demonstrating the benefit of spatial localization. Nevertheless, both SE-Block and CBAM operate on individual feature hierarchies and do not explicitly model interactions between multiscale representations. In contrast, the proposed MPFAN-AF achieves the highest performance, with an accuracy of 94.9% and an F1-score of 94.8%, by employing cross-branch attention fusion. This mechanism jointly optimizes complementary features extracted at different receptive fields, enabling more effective discrimination of visually similar disease classes and validating the proposed module's superiority over conventional attention blocks.

8. Conclusions and Future Directions

This research contributes to the development of the MPFAN-AF model, a significant technological innovation in plant disease classification. The model utilizes multiscale feature extraction, parallel feature aggregation, and attention fusion techniques. Consequently, the classification accuracy for tomato leaf disease is improved to 94.9%. In the agricultural domain, the MPFAN-AF model demonstrates strong real-time disease detection with high precision. The model is a synthesis of various aspects of complex disease patterns, in which multiscale attention and attention aggregation processes are crucial, focusing on the most pivotal regions to enable high-accuracy classification. In addition, MPFAN-AF outperforms more conventional DL-based models, such as ResNet-50, InceptionV3, and DenseNet-121, when a human classifier is present during disease recognition, thereby enhancing self-sustainability and output in crop disease management. The model's low parameter count and fast inference time, as demonstrated, provide a strong foundation for edge computing optimization. Future directions include hardware-specific deployment (e.g., on NVIDIA Jetson or ARM-based platforms), model quantization, and extensive field testing in varied environmental conditions to validate robustness and latency in true in-situ scenarios. The expectation of future advancements in AI, particularly in transfer and few-shot learning, will facilitate the development of advanced model capabilities for plant disease classification, enabling detection with few labeled examples. Moreover, multimodal data, such as environmental data and sensor data, would be a more accurate tool and would expand the scope of disease identification. Another potential avenue is the optimization of edge computing, which can be used to identify resources in resource-constrained environments. Eventually, when such sophisticated models are introduced into precision farming, they will improve disease control in crops, enable earlier treatment, and make practices more efficient and sustainable.

Article Information Form

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Ethical Approval: It is declared that during the preparation process of this study, scientific and ethical principles were followed, and all the studies benefited from are stated in the bibliography.

Conflict of Interest Notice The authors declare that there is no conflict of interest regarding the publication of this paper.

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